Code No: 111AL **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, September/October - 2021
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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JNTUH-USED SEPT-2021

1.a) Find the second difference of the polynomial $x^4-12x^3+42x^2-30x+9$ with interval of difference h=2.

b) Fit a curve y = a + bx by the method of least squares.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

[7+8]

2.a) Find y(3.4) from the following table using Newton's forward interpolation formula.

x	3	4	5	6
y	31	69	131	223

b) Fit a curve using Lagrange's interpolation formula.

x	0	5	10	15	20	25
y	12	15	17	22	24	30

[7+8]

3.a) Find a root of the equation $f(x)=x+\log x-2$ using Newton Raphson method.

b) Given that $\frac{dy}{dx}=1+xy$ and y(0)=1. Compute y(0.1) using Picards method upto three decimals. [7+8]

4. Find y(0.1) and y(0.2) using Runge Kutta method given that $y'=x^2-y,\ y(0)=1$. [15]

5. Find the Fourier sine transform of $\frac{1}{x\left(a^2+x^2\right)}$. [15]

6. Obtain a Fourier series for $f(x)=x\sin x$ in $(0\le x\le 2\pi)$. [15]

7. Solve the following partial differential equations:
a) $z^2(p^2+q^2+1)=c^2$
b) $x(y-z)\ p+y(z-x)q=z(x-y)$. [8+7]

8. Verify Green's theorem for $\int_c (3x-8y^2)dx+(4y-6xy)dy$ where C is the region bounded by $x=0,\ y=0$ *and* $x+y=1$. [15]

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